Kondor
10/06/2022

59

Roll No. ....

TCE–605

B. TECH. (CE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022

TRANSPORTATION ENGINEERING–I

(Highways and Airfield)

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define highway alignment. Describe the various requirements of highway alignment. (CO1)

   (b) Explain the following terms : (CO1)

   (i) Ruling gradient

   (ii) Limiting gradient

   (iii) Exceptional gradient

   (iv) Minimum gradient

   (v) Floating gradient

P. T. O.

(c) A highway is provided with a horizontal curve of radius 300 m in a certain locality. Calculate the superelevation required to maintain the design speed of 90 kmph. Calculate the maximum allowable speed. If the superelevation is limited to 0.07 and safe limit of transverse coefficient of friction is 0.15. (CO1)

2. (a) Differentiate between bitumen and tar. Discuss the various tests conducted of bitumen in road construction. (CO2)

(b) Explain the desirable properties of soil as a highway material. (CO2)

(c) List various tests to be carried on aggregate sample. Explain any *two* tests in detail. (CO2)

3. (a) Define traffic rotary. Discuss its advantages and limitations, in particular reference to traffic conditions in India. (CO3)

(b) Describe the objectives of origin and destination study. Explain various methods of origin and destination study. (CO3)

(c) Explain traffic capacity, basic capacity, possible capacity and practical capacity. (CO3)

The speed relationship for a particular road was found to be $u = (42.76 - 0.22k)$, where u is the speed in kmph and k is the density of vehicle per km. Find jam density, maximum capacity and density at maximum capacity.

4. (a) Differentiate between Flexible and Rigid pavements. (CO4)

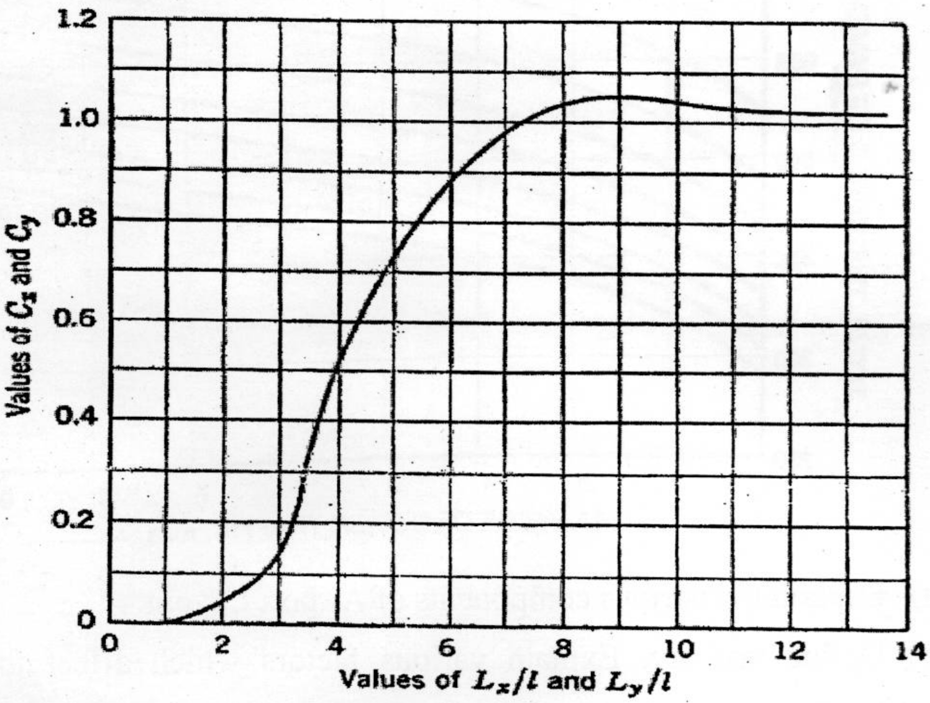
(b) Determine the warping stress at interior, edge and corner of a 25 cm thick cement concrete pavement with transverse joints at 11 m interval and longitudinal joints at 3.6 m intervals. The modules of subgrade reaction k is 6.9 kg/cm^3. Assume temperature differential during day to be 0.6°C per cm slab thickness. Assume radius of loaded area as 15 cm for computing warping stress at the corner. Additional data given below : (CO4)

$e = 10 \times 10^{-6}$ per °C

$E = 3 \times 10^5$ kg/cm^2

$\mu = 0.15$

Bradbury's Warping Stress Coefficients



(c) Design the pavement for construction of a new bypass with the following data : (CO4)

Two lane single carriage way

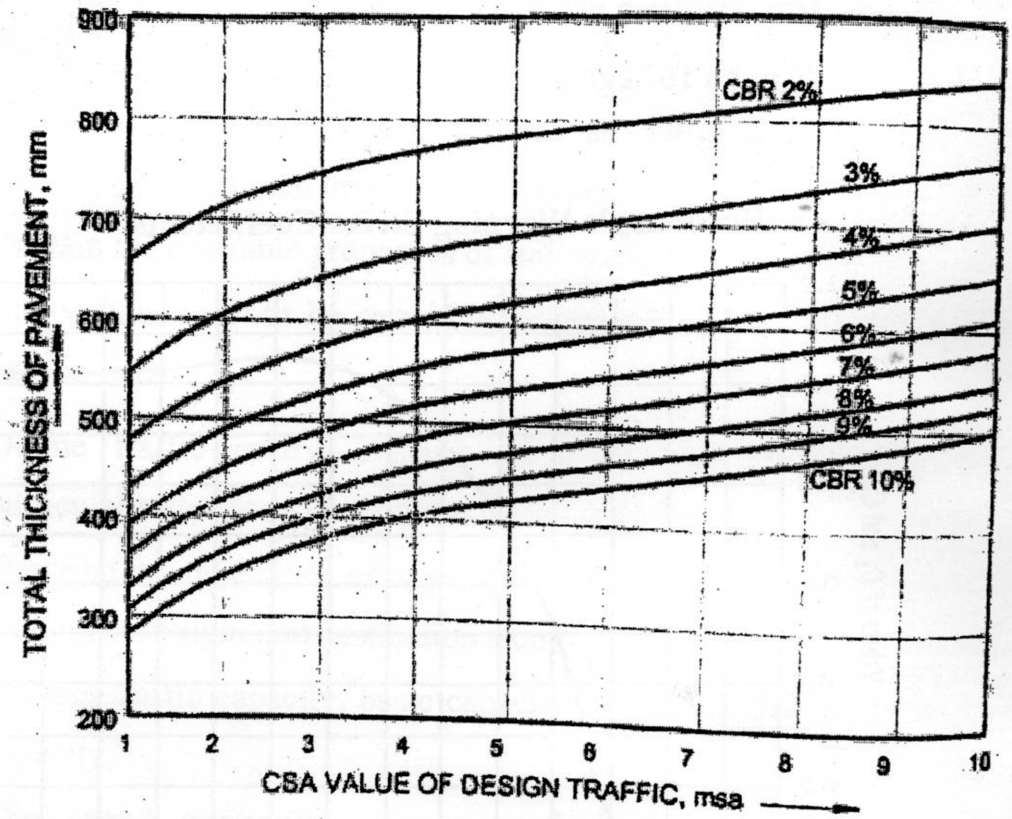
Initial traffic in a year of completion of construction work (sum of both directions) = 400 CVPD

Traffic growth rate per annum = 7.5 percent

Design life = 15 years

Vehicle damage factor = 2.5 (standard axles per commercial vehicle)

Design CBR value of sub-grade soil = 4%



5. (a) Explain the various components of Airport Layout. (CO5)

   (b) Define taxiway. Explain various factors which affect the layout of taxiway. (CO5)

   (c) The length of runway under standard conditions is 2100 m. The airport is to be provided at elevation of 410 m above mean sea level. The airport reference temperature is 32°C. If the runway is to be constructed with an effective gradient of 0.218 percent, determine the corrected runway length. (CO5)

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Roll No. ....

TCE–604

B. TECH. (CIVIL ENGINEERING) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022

QUANTITY ESTIMATION AND COSTING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Prepare a preliminary estimate of a polytechnic building for 800 students in order to assess the amount of fund on the following particulars : (CO1)

Carpet area required pr student = 1.20 sq. m

Area of corridor, verandah and lavatories etc. = 20% of the plinth area

Area of walls = 15% of the plinth area

Considered plinth area rate = ₹ 1,500.00 per sq. m

Cost of water supply = 6% of building cost

Cost of sanitation = 6% of building cost

P. T. O.

Cost of electrification = 11% of the building cost

Cost of approach road and boundary wall = 3.5% of the building cost

Contingency and work charged establishment shall be 5% and 2.5% of the total cost.

(b) Prepare a preliminary estimate of a building having plinth area equal to 1800 sq. m. Given that : (CO1)

(i) Plinth area rate ₹ 900 per sq. m

(ii) Extra for architectural work = 3.5 percent of the building cost

(iii) Extra for electrical installation (8%), water supply and sanitary installations (7%) of building cost.

(iv) Extra for other services 9% of the building cost

(c) Explain preliminary estimate and cube rate estimate. (CO1)

2. (a) Analyse the rate for 6 mm thick mosaic or terrazzo (1 : 1) layer over 20 mm thick cement concrete for 100 sq. m. Assume suitable data. (CO2)

(b) Explain Rate Analyse. What are the different factors affecting analysis of rate ? (CO2)

(c) Analysis the rate for 6 mm and 12 mm thick plastering work in ceiling for 100 sq. m area. Assume suitable data. (CO2)

3. (a) Figure shows a room of internal dimensions 4.0 m × 2.5 m, calculate the quantities of the following items of work by long wall and short wall method : (CO3)

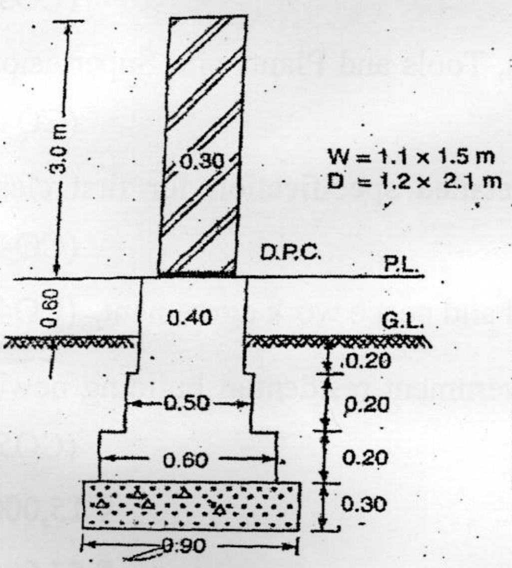
(i) Excavation in foundation

(ii) Lime concrete in foundation

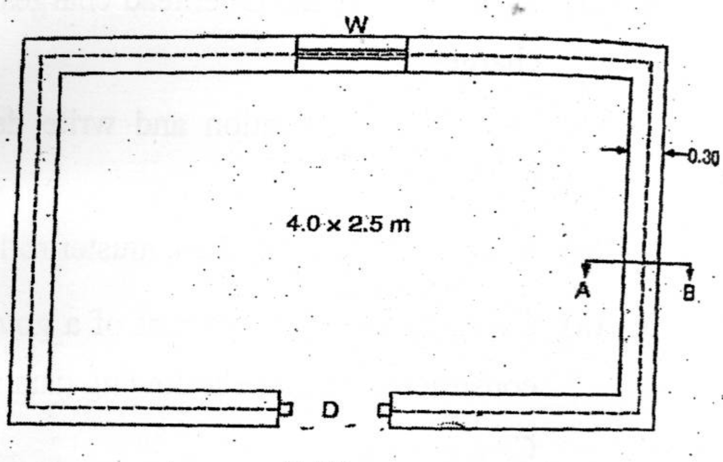
(iii) Brick work in cement mortar (1 : 4) in foundation and plinth

(iv) Brick work in cement mortar (1 : 6) in superstructure

(v) 2.5 cm thick DPC



Section at AB



PLAN

(All dimensions other than mentioned are in m)

(b) Reduced levels of ground along the center line of a proposed road from chainage from 0 to 200 are given below. The formation level at the 40 m chainage is 102.75. The formation of road from chainage 0 to 80 m has rising gradient of 1 in 40. The formation level has falling slope of 1 in 100 from chainage 100 to 200 m. The formation width of road at top is 12 m and side slopes of banking are 2 : 1. Draw longitudinal section of road and a typical cross-section and prepare estimate of each work for the road at a rate of ₹ 3.52 per m^3. (CO3)

Chainage	0	20 m	40 m	60 m	80 m	100 m	120 m	140 m	160 m	180 m	200 m
RL Ground	101.5	100.9	101.5	102	102.85	101.65	101.95	100.7	101.25	99.9	100.6
RL of Formation			102.75								
Gradient	← Rising gradient 1 in 40 →					← Falling gradient 1 in 100 →					

(c) Explain the terms Individual wall method and Centre line method. (CO3)

4. (a) Explain the terms Overhead charges, Tools and Plants and Supervision charges. (CO4)

(b) Describe specification and write detailed specification for first class building. (CO4)

(c) Explain bill of quantities, muster roll and peace work agreement. (CO4)

5. (a) Calculate the standard rent of a government residential building newly constructed from the following data : (CO5)

Cost of land ₹ 15,000

Cost of construction of the building ₹ 54,000

Cost of roads within compound and fencing ₹ 2,200

Cost of electric installation including fans 7% of the cost of the building.

Municipal house tax ₹ 600 per annum

Water tax ₹ 550 per annum

Property tax ₹ 380 per annum

(b) A three stories building on a plot of land measuring 850 sq. m. The plinth area of each storey is 70 sq. m. The building is of RCC framed structure and the future life may be taken as 70 years. The building fatches rent of ₹ 2,500 per month. Workout the capitalized value of the property on the basis of 5% net yield. For sinking fund 3% compound interest may assumed. Cost of land may be taken as ₹ 35 per sq. m. Other data required for working out the outgoings per annum assume suitably.

(CO5)

(c) Describe the purpose of valuation. Explain the different methods of valuation.

(CO5)

70

61

Roll No. ....

TCE–603

B. TECH. (CE) (SIXTH SEMESTER)

END SEMESTER EXAMINATION, 2022

WATER RESOURCES ENGINEERING–II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain briefly with neat sketchs the different forces that may act on a gravity dam. (CO1 and CO6)

   (b) How does the practical profile of a low gravity dam differ from that of the theoretical one and why ? (CO1 and CO6)

   (c) A concrete gravity dam has maximum reservoir level as 200.0 m, base level as 115.0 m, top and level of dam is 205.0 m. The upstream face is vertical and the downstream face batter starts at 199.0 m and is 1.0 H : 1.5 V. There is no tail water and consider full uplift pressure at heel.

Calculate : (CO1 and CO6)

(i) Maximum vertical stresses at heel and toe.

(ii) Major principal stress and shear stress near the toe.

(iii) Neglect earthquake and wave action.

2. (a) Draw a neat sketch (section) of zoned earth dam of 24 m height showing all parts and state the function in brief. (CO2 and CO6)

(b) Obtain the value of best central angle for constant angle arch dam layout. (CO2 and CO6)

(c) A homogeneous earthen dam 22.0 m high with top width 6.0 m, upstream slope = 3 : 1 and downstream slope = 2.5 : 1, with free board 2.0 m. Locate the phreatic line from the dam section and estimate the seepage flow through the body of the dam. Assume coefficient of permeability of material of the dam = 3×10^{-3} cm/sec. (CO2 and CO6)

3. (a) Enumerate the important types of spillway gates. Describe with a neat sketch the construction and working of Tainter gate. (CO3 and CO6)

(b) Explain the important relationship between tail water curve and jump height curve in the selection of energy dissipater. (CO3 and CO6)

(c) A high ogee spillway is to discharge 20 cumecs/m length of spillway. If the maximum water level is to be 50 m above river bed. Obtain the d/s profile of the spillway crest, using USWES method. Take the u/s face as vertical and d/s has slope of 0.8 H : 1.0 V. Assume suitable data if required. (CO3 and CO6)

4. (a) Draw a neat sketch of a typical layout of diversion head works and explain the function of its component parts. (CO4)

(b) Explain clearly with neat sketchs, the various corrections that are to be applied to the pressures at key points, as per Khosla's theory. (CO4)

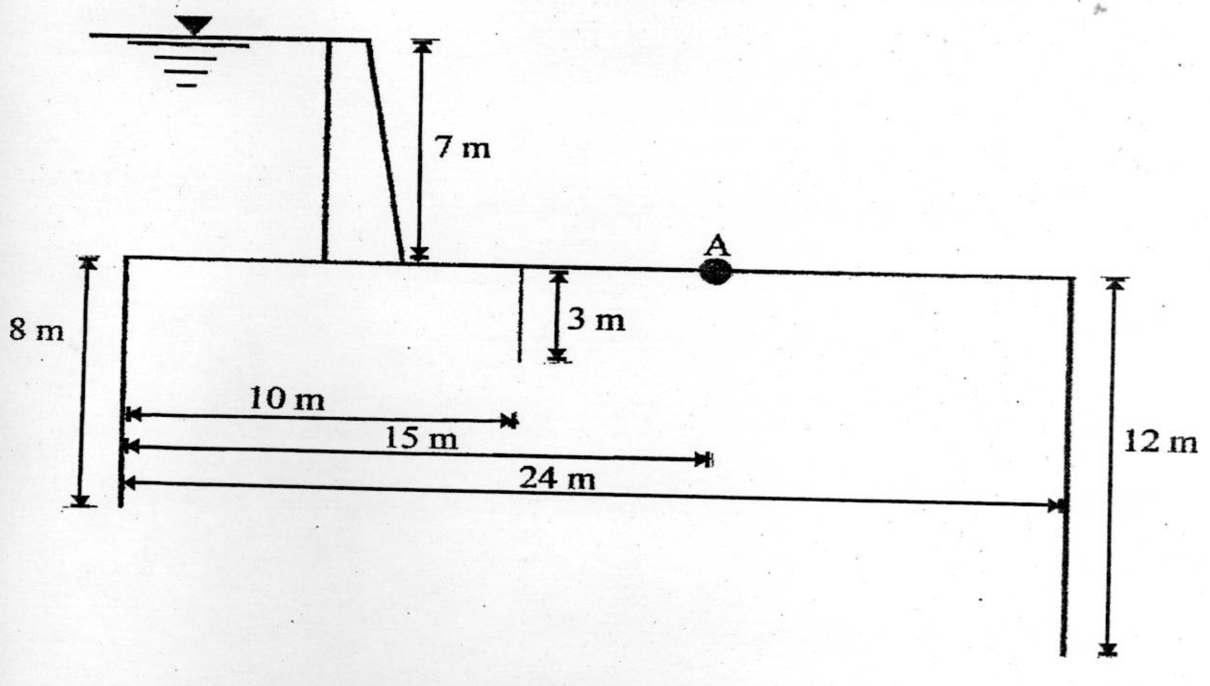
(c) The following figure shows the section of a hydraulic structure on a permeable foundation. Determine : (CO4)

(i) Average hydraulic gradient according to Bligh's creep theory

(ii) Uplift pressure at A

(iii) The floor thickness required at A

Take S = 2.24



5. (a) State various objects of river training. List out various methods of river training. Explain any *one* method with the help of a neat sketch. (CO5 and CO6)

(b) State the advantages and disadvantages of hydropower generation over thermal and nuclear power generation. (CO5 and CO6)

P. T. O.

(c) Determine the spacing of the tile drain from the following data :

(CO5 and CO6)

Average annual rainfall = 1200 mm

Discharge usually taken is 1% of average annual rainfall in 24 hours

Depth of impervious strata from land surface = 11 m

Highest position of water table below land surface = 1.5 m

Depth of drain = 1.8 m below land surface

Permeability of soil = 6×10^{-6} m/s

70

62 10

Roll No. ....

TCE–602

B. TECH. (CE) (SIXTH SEMESTER)

END SEMESTER EXAMINATION, 2022

REINFORCED CEMENT CONCRETE–II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain with proper reasoning various advantages and disadvantages of prestressed concrete design. Illustrate various methods of analysis of prestressed concrete member. (CO1)

(b) A symmetrical I shaped PSC beam given cross-section details of (top and bottom flanges 400 × 150 mm size, web thickness of 200 mm and the overall depth of beam is 600 mm) with 6 m long span is provided with 6-8 mm dia steel wires positioned at 75 mm from top fiber of the beam and 8 Nos. of 8 mm diameter steel wires whose resultant force is concentrated at 75 mm from the soffit of the beam. A net prestress of

P. T. O.

950 N/mm^2 is applied on all the wires; consider there is a net loss of 15% of loss in the prestress. Determine the stresses at top and bottom fibers of the given PSC beam at mid span and quarter span. Assume unit weight of the concrete as 24 kN/m^3. (CO1)

(c) Explain and derive equations for any *five* different types of losses in prestressing. (CO1)

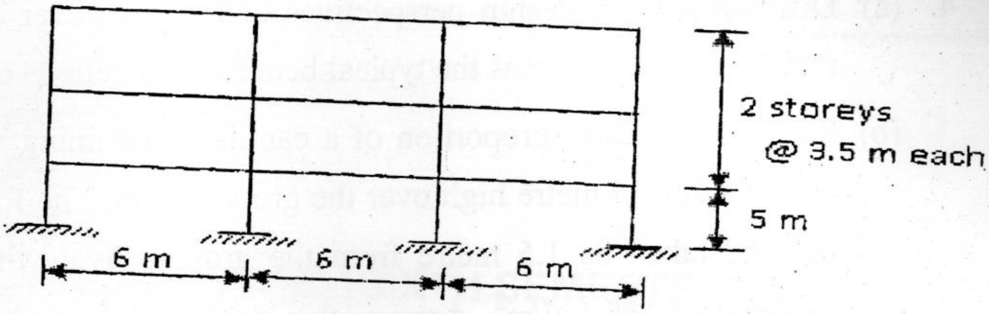
2. (a) State and explain your understanding with appropriate reasoning's behind the assumptions and guidelines made to apply moment redistribution for a statically indeterminate structure. (CO2)

(b) Draw the bending moment diagram of the two span continuous beam carrying characteristic live load of 20 kN/m in addition to its characteristic dead load of 5 kN/m. The length of the all span may be considered as 6 m. The bending moment diagram should be drawn for the maximum bending moment at the first intermediate support. Apply a 20% moment redistribution for the beam. (CO2)

(c) Demonstrate the percentage of length of shift of point of contra-flexure in a two span beam carrying a udl of '*w*' over its entire span. (CO2)

3. (a) Design a semi-circular (in plan) beam with radius 5 m is simply supported at ends and continuous over a column at its middle. The beam carries a uniformly distributed load of 25 kN/m throughout its length, including self-weight. Also draw shear force diagram, bending moment diagram and twisting moment diagram. (CO3)

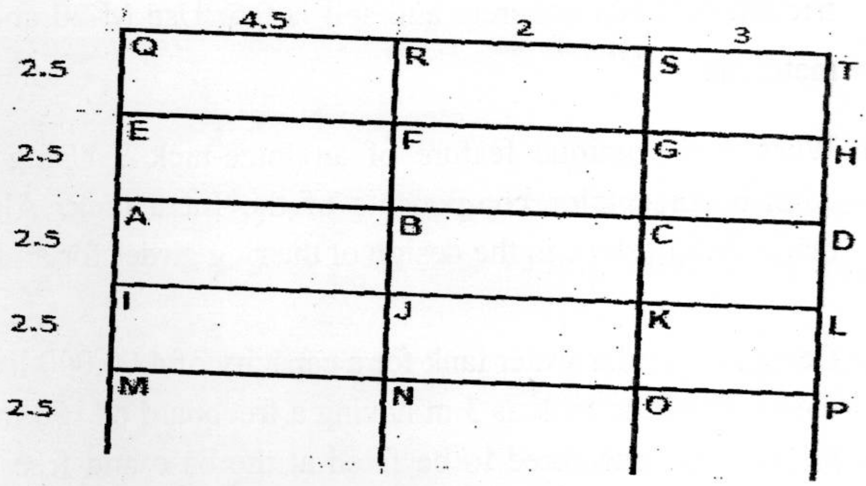
(b)



The structural framework for an office building is shown in the Figure. It has three storeys. There are 2 panels in one direction and three in the other. Each panel is 6.0 m wide between centers of columns, in both directions. The other data is as follows :

Live load on floor = 4 kN/m^2, Live load on roof = 1.5 kN/m^2, Beam = 3 50 mm × 500 mm, Column = 450 mm × 600 mm. Analyze the frame by using cantilever method. (CO3)

(c)



Analyze the member forces in the substitute frame ABCD selected , for Max Positive BM in the interior beam span, if the frame members have same section modulus and load details as given below. Member dead load of 12 kN/m and a live load of 18 kN/m run. (CO3)

4. (a) Differentiate the design perspectives of the cantilever and counterfort retaining walls. Discuss the typical benefits and deficits of both. (CO4)

   (b) Suggest a suitable proportion of a cantilever retaining wall to retain a level earth of 5 metre high over the ground level. The foundation depth may be taken as 1.5 metre from the ground level with safe bearing capacity of 180 kN/m^2. Assume that the backfill has a unit weight of 16 kN/m^3 and an angle of repose of 30°. Further, assume a coefficient of friction between soil and the base slab as 0.53. Design the base slab (heel slab) or stem slab. (CO4)

   (c) Design a counterfort retaining wall to retain 7 m high embankment above ground level. The foundation has to be taken 1 m deep. Soil SBC is 180 kN/m^2. Top of earth retained is horizontal and soil unit weight is 18 kN/m^3 with an angle of internal friction 30° and coefficient of friction between concrete and soil is 0.5. Use M-20 and Fe-415 grade materials. (CO4)

5. (a) What is the unique feature of an Intze-tank ? Using a neat sketch, describe the major components of the Intze tank. Also, discuss the major design steps in the design of the ring girder for an Intze tank. (CO5)

   (b) Design a circular water tank for a capacity of 4,00,000 litres considering the depth of the tank as 3 m having a freeboard of 150 mm. The wall of the tank is considered to be fixed at the base and free at the top. Use M-25 and Fe-415 steel. (CO5)

   (c) Discuss the design procedure for the supporting structure of overhead circular water tank. (CO5)

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63

Roll No. ....

TCE–601

B. TECH. (CE) (SIXTH SEMESTER)

END SEMESTER EXAMINATION, 2022

ENVIRONMENTAL ENGINEERING–II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe and explain the time variation in sewage flow and their effects on the design of sewer capacities. (CO1)

(b) BOD of a sewage incubated for 2 days at 30°C was found to be 160 mg/L. Find the value of 5-day 20°C BOD. Assume K (base 10) at 20°C as 0.12 per day. (CO1)

(c) Write in detail about the characteristics of sewage. (CO1)

P. T. O.

2. (a) Find the combined flow discharge of sewage for the given data : Area to be served is 150 hectare, Population density is 50000, Time of entry is 5 minutes, Time of flow is 20 minutes. Rate of water supply is 135 lpcd, Impermeability factor = 0.45.

   Assume 80% of water supplied turns into sewer and peak factor as 1.5. (CO2)

   (b) Explain the process of testing of newly laid sewer lines before bringing them into commission. (CO2)

   (c) Describe the one pipe and two pipe plumbing systems. Compare them. (CO2)

3. (a) Describe about the component parts of septic tank, its advantages and disadvantages with neat sketches. (CO3)

   (b) A grit chamber is designed to remove particles with a diameter of 0.2 mm, Specific gravity = 2.65. Settling velocity for these particles has been found to be range from 0.016 to 0.22 m/s, depending on the shape factor, a flow through velocity of 0.3 m/s will be maintained by proportioning weir, determine the channel dimensions for a maximum waste water flow of 10,000 cum/day. (CO3)

   (c) State the objectives of primary treatment. Discuss about the grit chambers which is adopted in sewage treatment. (CO3)

4. (a) Describe with neat sketches about the trickling filters. (CO4)

   (b) Determine the size of standard rate trickling filter to treat 6 million litres of sewage per day having BOD of 160 mg/L. Take hydraulic loading of 6 $m^3/m^2/d$ and organic loading of 0.35 $kg/m^3/d$. (CO4)

   (c) Compare the advantages and disadvantages of ASP and Trickling Filters. (CO4)

5. (a) What is sewage farming ? List its methods and state its advantages over the method of disposal of sewage dilution. (CO5, CO6)

(b) Write short notes on the following : (CO5, CO6)

(i) Waste water reclamation

(ii) Sewage disposal to sea water.

(c) A waste water treatment plant produces sludge of 1000 kg dry solids per day with a moisture content of 97%. The solids are 65% volatile with specific gravity 1.05 and inorganic solids of specific gravity 2.55.

Determine the sludge volume of raw sludge, after dewatering to 70% and after incineration. (CO5, CO6)

70